

CAMERA CONTROLLER – signal flow

GND is a global net on every sheet. Double-click a block to open it.

1. Battery 2S + protection



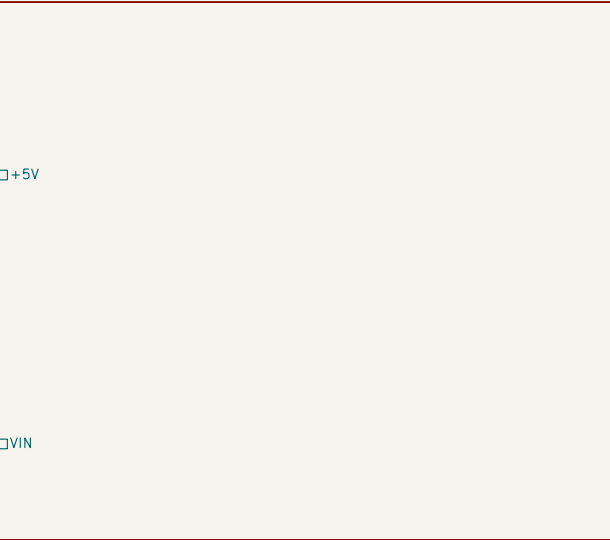
File: sheets/power_in.kicad_sch

2. Buck to 5 V



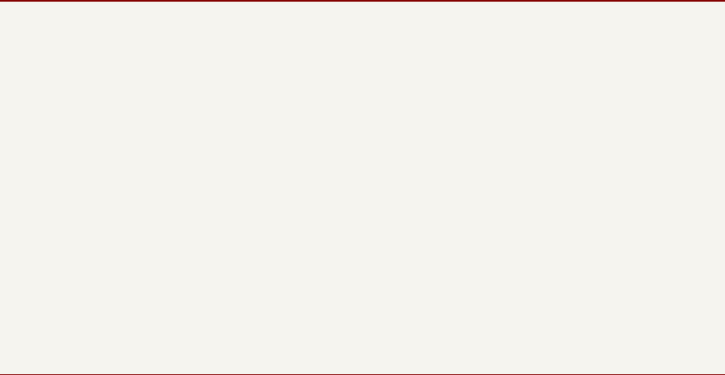
File: sheets/buck_5v.kicad_sch

3. ESP32 + camera



File: sheets/esp32_camera.kicad_sch

4. COTS / SRAD supplies



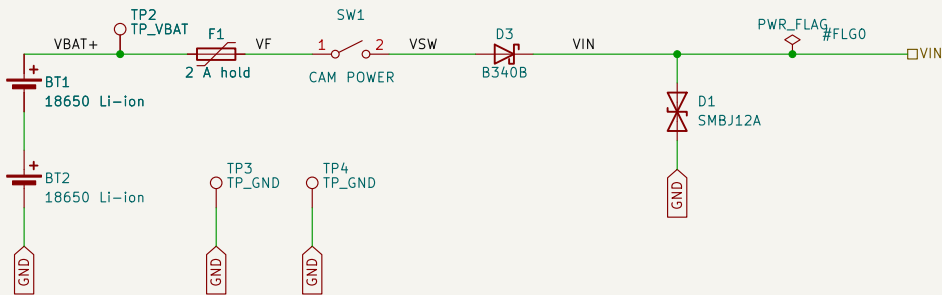
File: sheets/supplies.kicad_sch

Block 4 is electrically isolated from the rest of this board on purpose.

Stanford SSI			
Sheet: /			
File: battery_pcb.kicad_sch			
Title: Battery PCB – AV bay power, arming and camera			
Size: A3	Date:		Rev: A
KiCad E.D.A. 10.0.0	Id: 1/5		

2 x 18650 in series: 7.4 V nominal, 8.4 V full. Charge the cells off the board, one at a time, in a Li-ion charger.

F1: 2 A PTC right at the pack +. D3: reverse-polarity Schottky (B340B, about 0.35 V at the 0.6 A load).
D1: TVS, 12 V standoff (above the 8.4 V full pack), clamps below 20 V; everything on VIN is rated 25 V or more.
Low-battery cut-off: U1's EN divider on sheet 2 (about 2.9 V per cell).



Stanford SSI		
Sheet: /1. Battery 2S + protection/		
File: power_in.kicad_sch		
Title: 1. Camera + ESP32 supply (2S)		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0		Id: 2/5

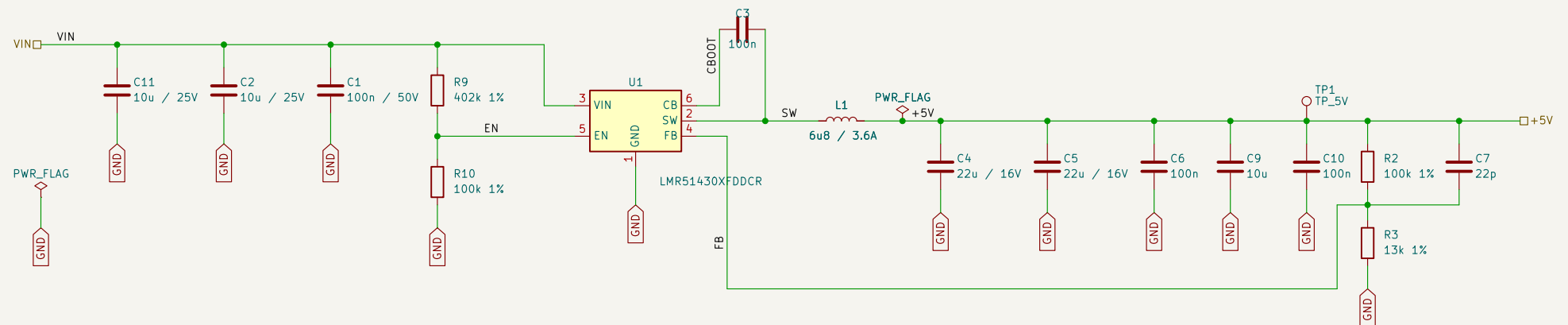
LMR51430 buck, 500 kHz: $V_{OUT} = 0.6 \times (1 + 100k / 13k) = 5.22 \text{ V}$

Input caps: C1 (100 nF) sits right at U1's VIN/GND pins; C2, C11 (10 uF) are the bulk.

UVLO (R9/R10 on EN): on above $1.23 \text{ V} \times (1 + 402k/100k) = 6.2 \text{ V}$, off below $1.08 \text{ V} \times 5.02 = 5.4 \text{ V}$ at VIN,

so the camera pack stops at about 2.85 V per cell instead of being run flat (U1's own UVLO is 3.6 V).

Output caps: C4, C5 (22 uF) and C6 (100 nF) sit at L1 / U1. C9 (10 uF) and C10 (100 nF) sit at the DevKIT's 5V pin (J1 pin 19) on the board.



Stanford SSI

Sheet: /2. Buck to 5 V/

File: buck_5v.kicad_sch

Title: 2. Buck 8.4 V -> 5.0 V

Size: A4

Date:

KiCad E.D.A. 10.0.0

Rev: A

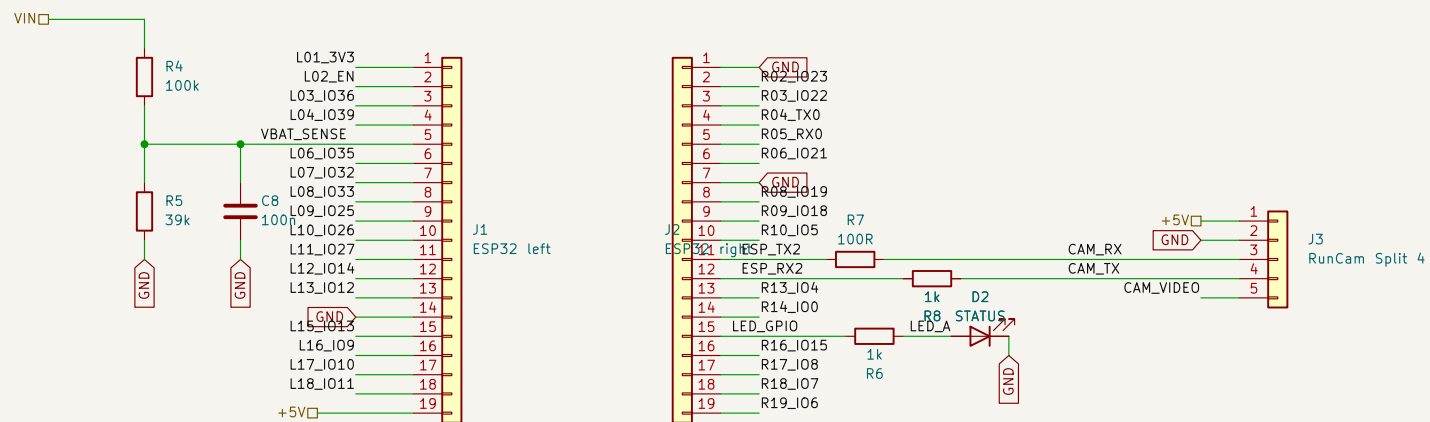
Id: 3/5

2 x 1x19 female header for the ESP32-DevKitC V4. GPIO6-11 go to the module SPI flash – do not use.

Pack sense: $V_{BAT_SENSE} = V_{IN} \times 39k / (100k + 39k)$. Full pack 8.4 V (V_{IN} about 8.1 V after D3) $\rightarrow$ 2.26 V at GPIO34, inside the ESP32 ADC's 2.45 V range (11 dB).

Firmware: $V_{IN} = V_{BAT_SENSE} \times 139 / 39$; pack = $V_{IN} +$ about 0.35 V.

RunCam Split 4: 5-20 V In. Single 5-pin harness: power, ground and UART (R7 / R8 in series with the UART lines).



Stanford SSI

Sheet: /3. ESP32 + camera/

File: esp32_camera.kicad_sch

Title: 3. ESP32-DevKitC V4 socket + RunCam Split 4

Size: A4

Date:

Rev: A

KiCad E.D.A. 10.0.0

Id: 4/5

Own cell, own switch, per DTEG 6.9.1. SW2 / SW3 take the pin-switch leads; they sit next to the pin-switch stack. All connectors latch (Micro-Fit 3.0).

These returns are deliberately NOT tied to GND – that independence is the point.

